

SYLLABUS :-

Development of Mineral Deposits: x Introduction to rock, minerals ores and deposits. x Shape and sizes of ore and coal deposits; Coal as energy minerals x Mining cycles and unit operations: Mineral prospecting and exploration, Mine evaluation and Investment, Mine development and Exploitation, unit operations: mines-to mill x Rock fragmentation: i Machine cutting: various cutting tools, methodologies and numerical calculations, ii Drilling and Blasting: drilling machines and fluids, methods and calculations, explosives and their properties, fragmentation by blasting technology. x Conventional and special methods of shaft sinking, shaft lining, deepening and widening of shafts; Raising and winzing. x Access and egresses to ore deposits; shape and size vis a-vis machinery and production. x Operations in quarry, opencast and underground mines. 3-0-0 Text Books: 1. Howard L. Hartman, Jan M. Mutmanský 2007 Introductory Mining Technology, 2nd Edition, John Wiley and Sons. 2. Peter Darling Edited 2011 SME handbook Vol 1 and 2, 3rd Edition, SME publications. 3. R. D. Singh 1997 Principles and Practices of Modern Coal Mining, New age International P Ltd. Publishers. 4. R. R. Tatiya 2013 Surface and Underground Excavations, 2nd Edition: Methods, Techniques and Equipment, CRC Press. 5. S. K. Das 2014 Explosive and Blasting Practices in Mines, Lovely Publication. 6. S. K. Das 1994 Modern Coal Mining Technology, Lovely Publications. 7. W. A. Hustrulid, M. Kuchta, K. Martin 2013 Open Pit Mine Planning and Design, Vol 1 and 2, 3rd Edition, CRC Press